3.2 Exponents and Powers



We can write large numbers in simplified form as given below.

For example, $16 = 8 \times 2 = 4 \times 2 \times 2 = 2 \times 2 \times 2 \times 2$

Instead of writing the factor 2 repeatedly 4 times, we can simply write it as 2^4. It can be read as 2 raised to the power of 4 or 2 to the power of 4 or simply 2 power 4.

This method of representing a number is called the exponential form. We say 2 is the base and 4 is the exponent.

Note

The exponent is usually written at the top right corner of the base and smaller in size when compared to the base.

Let us look at some more examples,

$64 = 4 \times 4 \times 4 = 4^3$ (base is 4 and exponent is 3)

Also, $64 = 8 \times 8 = 8^2$ (base is 8 and exponent is 2)

$243 = 3 \times 3 \times 3 \times 3 \times 3 = 3^5$ (base is 3 and exponent is 5)

$125 = 5 \times 5 \times 5 = 5^3$ (base is 5 and exponent is 3)

Remember, when a number is expressed as a product of factors and when the factors are repeated, then it can be expressed in the exponential form. The repeated factor will be the base and the number of times the factor repeats will be its exponent.

We can extend this notation to negative integers also. For example,

$-125 = (-5) \times (-5) \times (-5) = (-5)^3$ [base is '-5' and exponent is 3]

Hence, $(-5)^3$ is the exponential form of -125.

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In a calculator, multiplication of two large numbers is displayed as 1.219326E17 which means 1.219326×10^{17} where, 'E' stands for exponent with base 10.



Numbers in Exponential Form

Now, we will see how to express numbers in exponential form.

Let us take any integer as 'a'.

Then, $a = a^1$ ['a power 1']

$a \times a = a^2$ ['a power 2' ; 'a' is multiplied by itself 2 times]

$a \times a \times a = a^3$ ['a power 3' ; 'a' is multiplied by itself 3 times]

$\vdots \quad \vdots \quad \vdots$

$a \times a \times \ldots \times a (n \text{ times}) = a^n$ ['a power n' ; 'a' multiplied by itself n times]

Thus we can generalize the exponential form as a^n, where the exponent is a positive integer $(n > 0)$.

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